L3 — Linear Probe (Logistic Regression)

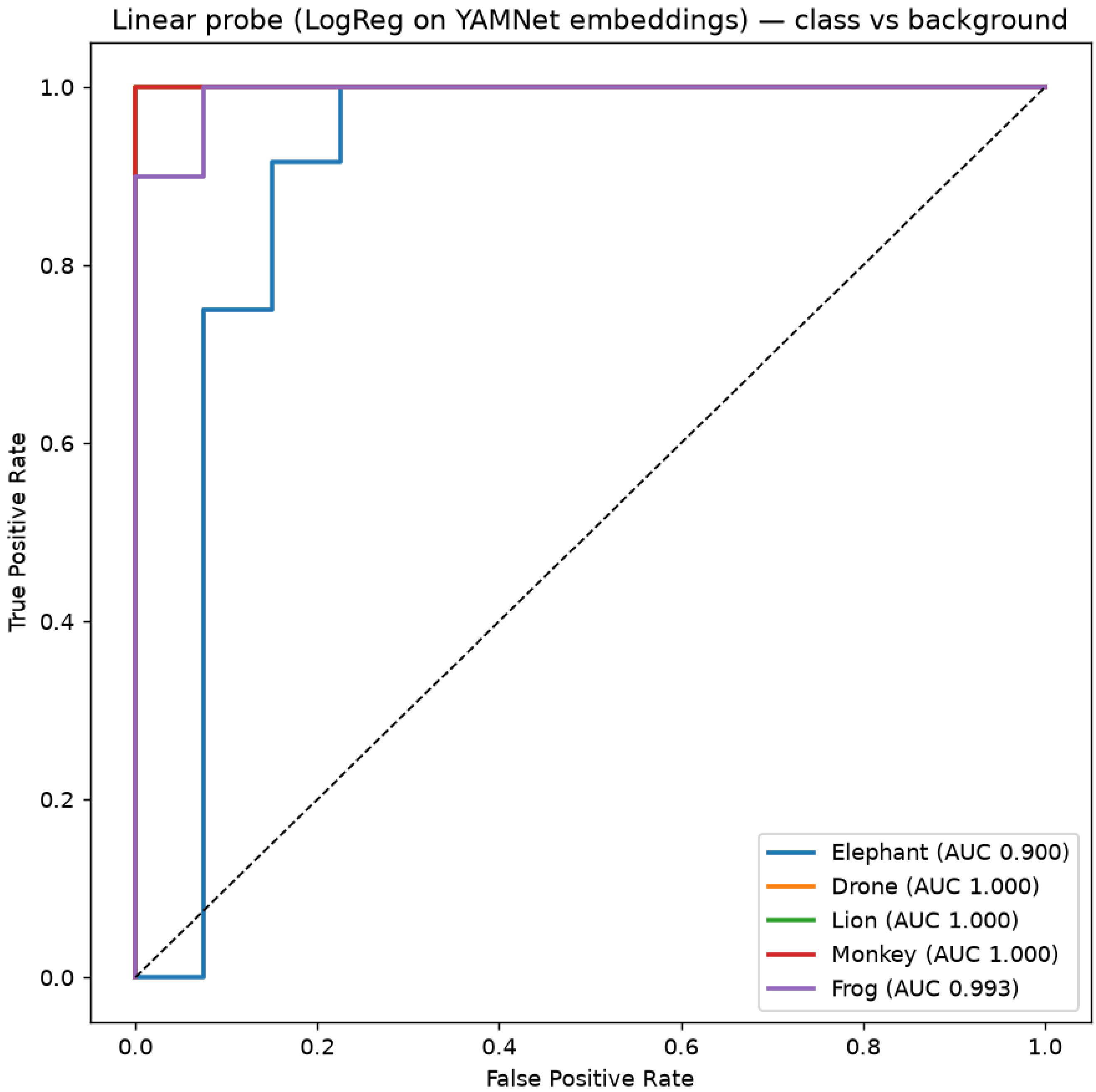
plan.md leg: "Simple Classifier: Linear Probe". **Goal:** the ceiling for a frozen backbone with zero complexity. **Model:** YAMNet (frozen) → mean-pool → LogisticRegression, one binary classifier per class vs background. **Script:** `experiments/scripts/leg3_linear_probe.py`. **Artifacts:** `figures/roc_linear_probe.png`, `experiments/outputs/leg3_linear_probe.json`.

Method

Per target class: standardize embeddings, fit `LogisticRegression(class_weight='balanced')` on the **train** split, evaluate on the **test** split. Splits are the source-level 80/20 from `01_data_prep.md` (no window leakage). Background is the negative for every class.

Finding: near-perfect, linearly separable (Elephant the only soft spot)

class	AUC	Acc	Precision	Recall	F1	pass (AUC>0.70)
Elephant	0.900	0.846	0.625	0.833	0.714	✅
Drone	1.000	1.000	1.000	1.000	1.000	✅
Lion	1.000	1.000	1.000	1.000	1.000	✅
Monkey	1.000	1.000	1.000	1.000	1.000	✅
Frog	0.993	0.980	1.000	0.900	0.947	✅



- **All five PASS** the AUC > 0.70 criterion, four of them at/near 1.0.
- **Elephant** is the weakest: AUC 0.90 with **precision 0.625** — it recalls elephants well (0.83) but picks up background false positives, exactly the embedding overlap seen in L2.
- This ROC is the **canonical baseline** every later experiment is compared against; ROC points are saved in `leg3_linear_probe.json` for overlaying.

Interpretation

A plain linear classifier on frozen YAMNet embeddings is essentially at ceiling. The embedding space is **not** the bottleneck (refuting any need for backbone surgery on this clean data). The only headroom is Elephant-vs-background precision, which is a *data/feature separability* issue (low-freq rumble ≈ ambient), not a model-capacity one.

Pass criterion (AUC > 0.70): ✅ all classes. Embedding space confirmed correct.

> ⚠ **Caveat for the bigger plan:** these are clean library clips. The `experiments.pdf` program trains/evaluates on **post-ODAS** reconstructed spectra (beamformed, Griffin-Lim artefacts, ghost-track false positives) — much harder. This near-perfect probe is the *upper bound*, not a deployment estimate.